Prithish Saha

+91 87777 59829 | prithish.saha44@gmail.com | LinkedIn | GitHub | LeetCode

SUMMARY

Final-year B.Tech Computer Science student with ML internship experience at WEBEL (Govt. of West Bengal) and Jadavpur University. Experienced in building full-stack AI applications, computer vision systems, and LLM-powered pipelines. Passionate about software engineering and AI innovation, with a rapid-learning mindset and enthusiasm for exploring new technologies. Eager to solve real-world problems by building scalable and production-ready AI systems.

Education

MCKV Institute of Engineering — Howrah, West Bengal
B.Tech in Computer Science & Engineering — YGPA: 9.5 (Y1), 8.89 (Y2) — *Aug. 2023 – June 2027*

Sinthee Ramkrishna Sangha Vidyamandir — Kolkata, West Bengal
Higher Secondary (Science) — WBCHSE — Score: 85.8% — *2021 – 2023*

Sinthee Ramkrishna Sangha Vidyamandir — Kolkata, West Bengal
Secondary — WBCSE — *2021 – 2023*

Experience

Machine Learning Intern — March 2026 – June 2026
WEBEL – Centre of Excellence on DS & ML, Govt. of West Bengal — *Kolkata, West Bengal*

- Developed a privacy-preserving handwritten prescription reading and digitization pipeline using FastAPI and Gemini 3.5 Flash for structured JSON extraction.
- Trained and deployed a custom YOLOv8n model for automated PII detection and anonymization of handwritten prescription.
- Designed a dual-layer Actor–Critic evaluation framework combining rule-based validation and AI semantic auditing to detect extraction errors and hallucinations.

Summer Intern — June 2025 – Aug. 2025
Jadavpur University — *Kolkata, West Bengal*

- Curated and processed a comprehensive custom dataset of diverse environmental waste and garbage images to build a robust foundation for computer vision pipelines.
- Annotated over 300 complex images with precise bounding boxes and class labels, establishing a high-quality ground-truth dataset required for accurate object localization.
- Integrated Large Language Models (LLMs) with the LangChain framework to classify complex waste compositions and autonomously compute environmental severity scores.

Projects

Waste Watch | *Python, Flask, React, YOLO, Leaflet.js, LLM*

- Built an AI-powered web application for garbage detection and environmental risk assessment.
- Integrated YOLO for waste localization and enabled users to select garbage locations via interactive maps using Leaflet.js.
- Applied LLM-based analysis to estimate waste composition and designed a weighted severity scoring model.

AI Story Generator Web App | *Python, Streamlit, FastAPI, Google Gemini API, gTTS*

- Developed a full-stack AI web application that generates creative stories from user prompts using Google Gemini AI API.
- Implemented FastAPI-based backend services with Streamlit frontend for real-time prompt processing.
- Integrated text-to-speech pipeline using gTTS to produce automated spoken audio outputs.

Technical Skills

Languages: C/C++, Python, Java, SQL, JavaScript, HTML/CSS
Frameworks: React, Node.js, Flask, FastAPI
Machine Learning & AI: Deep Learning, Computer Vision, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks
Databases: MongoDB, MySQL, ChromaDB
Developer Tools: Git, GitHub, Docker, AWS, VS Code, Tableau, Jupyter Notebook, Roboflow
Libraries: NumPy, Pandas, TensorFlow, OpenCV, Scikit-learn, YOLO